

Accepted Service Adaptation: Finite Continuation Prices and the Value of Memory

Anonymous manuscript for Operations Research

Abstract. A service provider with renewal participation constraints must decide how much past information to retain. We identify a class of continuous-tier contracts in which exact optimal adaptation can be constructed on a recombining public graph without enumerating histories. Positive additive payments, quadratic operating rewards, and the absence of intertemporal switching make each continuation cap introduce at most one new payment-price threshold. A finite global breakpoint set yields an explicit arithmetic complexity bound, while the optimizer retains only the largest threshold encountered along the realized history. We prove a matching linear-order lower bound on the number of necessary memory labels and derive an exact memory-value frontier for a family of renewal agreements. A graph-sized quadratic program optimizes a shared zero-release table; at positive release, the finite-price construction supplies exact fixed-table values and supporting cuts. The broader switching, release-rent, promise-state, and deployment results are retained with their proofs. Exact coupled-graph experiments extend to 64 dates and include independent tree and shared-table checks. Earlier unfavorable cache evidence remains explicit: 83 of 84 strict-tolerance queries refreshed, without a measured cache speed advantage.

Keywords: continuation participation; contractual memory; stochastic optimization.

Subject classifications: Dynamic programming/optimal control: service adaptation; Programming: stochastic models; Inventory/production: service contracts.

Area of review: Stochastic Models.

1 Introduction

A service provider can improve its operating policy and still be unable to offer it. Customers with renewal or cancellation rights evaluate what the agreement promises from the current review onward. The provider must therefore respect participation at each continuation, not only the surplus forecast when the agreement was signed. A second constraint is organizational: a service rule may use a preannounced schedule, the current operating regime, a small memory record, or the entire observed history. The value of adaptation must compare the best accepted contracts under the competing information architectures.

This paper identifies a class in which the information needed for an optimal accepted contract can be constructed exactly without enumerating histories. The class has public exogenous Markov uncertainty, continuous service tiers, quadratic operating rewards, positive additive payments, and continuation payment caps. For the principal computational theorem, changing a tier carries no intertemporal switching cost. Participation constraints may bind at any review; their cumulative effect is the central feature, not a small perturbation.

The key object is the payment response to a continuation price. A local service decision has two tier thresholds. At each public state, imposing its continuation cap adds at most one further threshold. Backward propagation therefore produces a finite global set of price breakpoints whose size grows linearly with the public graph, rather than with its exponentially larger history tree. The provider executes the optimizer by retaining the largest participation-price threshold encountered so far. This gives an explicit exact algorithm, a bounded state representation, and independently auditable primal–dual equality. No full-tree optimum or full-tree dual solution is assumed available to generate the representation.

The price record has operational content. Two customers reaching the same public regime may require different service tiers because earlier renewal constraints imposed different effective prices. We prove that the worst-case number of distinguishable price labels grows linearly with the public graph. For a family of renewal agreements, we derive the exact value attainable with a limited number of memory symbols. The optimal encoding groups ordered continuation caps into contiguous cells, and a finite dynamic program computes the resulting value–memory frontier. This specifies what a memory device produces before any installation cost is assigned to it.

A shared policy table remains a global commitment. We derive a graph-sized quadratic program for its optimized zero-release value, including an extension with switching costs. At positive release, the finite-price algorithm supplies exact fixed-table values and supporting prices for a Benders-style shared-design master. The outer master is standard, and no universal finite exact cut count is inferred. The constructive finite-price result is distinguished throughout from the more general extensive-form-dual representation retained for switching and promise-state models.

The paper keeps the broader mathematical framework rather than replacing it with the tractable subclass. Print appendices retain the release frontier, comparator-adjusted continuation rents, friction paths, general promise recursion, shared-table representation, and deployment certificates. Their complete supporting proofs remain in the electronic companion. These results describe accepted adaptation beyond the principal finite-price theorem; they are not additional claims of new decomposition principles.

Exact experiments stress horizon length, public-state recombination, binding continuations, fixed shared

corridors, and optimized tables. They include independent unfolded-tree matrix checks and a separate numerical optimizer on small instances. No field data or calibration are claimed. The earlier adverse computational findings remain visible: in the inherited strict-tolerance cache experiment, 83 of 84 queries refreshed and the cached pipeline was not faster. That experiment is not repurposed as evidence for the new algorithm. The new contribution is the constructive finite-price theorem and its memory consequences, supported by exact coupled-graph computations.

2 Relation to Dynamic Contracting and Optimization

Continuation states are established in dynamic contracting. Spear and Srivastava (1987) and Thomas and Worrall (1988) use recursive contractual promises in environments with incentive or enforcement constraints. In Operations Research, Chen et al. (2020) study optimal monitoring when unobserved effort changes the arrival of adverse events. Their formulation uses the agent’s promised continuation utility as a state, with promise keeping, incentive compatibility, and continued participation determining its evolution. Liang et al. (2023) study resource-allocation contracts that induce effort to reduce adverse events and derive implementable allocation structures using dynamic incentives and self-generating sets.

Our public-state institution does not solve those information problems. Its controls are contingent service tiers under externally fixed prices, and its binding constraints are continuation participation and physical or payment commitments. The question is which accepted deviations improve upon an already optimized information restriction. A *continuation price* here is a multiplier on a continuation participation constraint. It is not the customer’s promised utility. The promised payment in Section C is instead a primal state variable. This distinction prevents a dual representation from being mistaken for a new promised-utility construction.

The journal-local boundary also includes maintenance, capacity partnerships, and dynamic assignment. Tian et al. (2021) choose payments and potential termination to induce unobserved maintenance and repair effort. Liu et al. (2019) design renegotiated membership agreements and transfers under privately updated demand information. He (2023) derives deadline mechanisms for project assignment with a privately evolving cost type and limited capacity. Here demand and service are public, prices are fixed, and customer exit is a participation constraint rather than a privately informed report or an endogenous termination instrument. Switching friction enters the provider’s operating objective. The structural question is the value of relaxing a specified public-information restriction, and its implementation through an exact remaining-payment state; we do not claim a solution to those incentive-design problems.

Constrained Markov decision processes supply resource-state and occupation-measure methods (Altman, 1999). Strong convexity, normal cones, and sensitivity of convex values are standard (Boyd and Vandenberghe, 2004). Absolute-value switching is an instance of a generalized-lasso penalty (Tibshirani and Taylor, 2011). Explicit parametric quadratic programming supplies the piecewise-affine policy and piecewise-quadratic value structure used below (Bemporad et al., 2002). We do not claim any of these general principles as new. Our use of them distinguishes restriction rents from participation rents, identifies the continuation-slack event that terminates a profitable release, and accounts for reoptimization of the comparator. The transfer identity by itself establishes none of those value comparisons.

Robust set containment is also an established device (Ben-Tal and Nemirovski, 2000; Bertsimas and Sim, 2004). We use it only for certification and quantify its local loss. Numerical optimizers, including OSQP (Stellato et al., 2020), produce candidate decisions and prices. The evidence below accepts a decision only after a separate rational calculation has checked feasibility and its primal–dual bracket. The historical value-gradient and learned-price results remain in the accompanying research archive, with their assumptions and proofs intact; they are implementation results rather than the identity of this paper.

Modern multistage decomposition is the closest computational comparison. Girardeau et al. (2015) prove almost-sure convergence for a class of sampling-based nested decomposition algorithms with general convex stage costs. Dowson (2020) formulate policy graphs and extend stochastic dual dynamic programming to a class including convex infinite-horizon problems with continuous states and controls. Füllner and Rebennack (2025) survey the assumptions, variants, and enhancements of stochastic dual dynamic programming. Lan (2023) analyze its iteration complexity. These results are substantially broader than our tractable contract subclass. Merely choosing a shared design variable, adding a promise state, or normalizing a continuation multiplier is not a distinction from that literature.

The residual conclusion here is different: scalar continuation caps act by truncating a monotone payment response, each cap contributes at most one new breakpoint, and a single finite union suffices across all vertices of a recombining public input graph. The construction generates those thresholds before the root optimum is known. It gives an arithmetic storage and construction bound, and identifies the changing information record needed for execution. It does not assert that the number of generic Benders cuts or the bit cost of exact arithmetic is uniformly small.

Separable resource allocation is also a close comparison, not an omitted alternative. Vidal et al. (2016) develop decomposition for nested resource allocation; Schoot Uiterkamp et al. (2021) give a fast algorithm for quadratic allocation with nested constraints; Wu et al. (2021) develop an exact combinatorial method for separable convex allocation with nested bound constraints. Their resource-allocation algorithms make monotone prices and threshold operations established tools. We do not claim a new water-filling principle. The result proved here measures complexity in the compact stochastic public graph, retains different inherited prices when histories recombine, and derives the exact memory-value frontier for the stated renewal institution. Applying an algorithm to an already unfolded resource-allocation instance does not by itself give that compact-input bound. The full-tree and graph-sized descriptions are explicitly distinguished in the proof.

Nonanticipativity and policy aggregation (Rockafellar and Wets, 1991), shared-design Benders masters (Benders, 1962), and stochastic dual value approximations (Pereira and Pinto, 1991) remain the appropriate descriptions of the general representation in the print appendices. The finite-price theorem supplies a constructive stronger conclusion for a specified subclass, not a renamed version of those general objects. The accompanying result-by-result comparison records both the inherited principles and the precise additional claim being submitted for scrutiny.

3 A Service Agreement with Continuation Participation

3.1 Primitives and accepted continuations

At date $t = 0, \dots, T - 1$, a provider observes a public regime z_t and an inherited tier q_t . It chooses stock S_t and a tier $x_t \in [0, 1]$ before demand is realized. Demand D_z is exogenous, nonnegative, and has finite mean. Regime transitions do not depend on the policy. The next inherited tier equals the chosen tier. Premium $r_z x$, indemnity $\kappa x(D_z - S)^+$, and physical cost parameters are fixed in the agreement. There is no private information. With $f_z(S) = \mathbb{E} \min(S, D_z)$ and $L_z(S) = \mathbb{E}(D_z - S)^+$, the customer's expected stage utility is

$$u(z, S, x) = \nu f_z(S) - a(z, S)x, \quad a(z, S) = r_z - \kappa L_z(S), \quad \nu > 0. \quad (1)$$

Maintenance and switching costs are resources expended by the provider, not transfers received by the customer.

An operational interpretation is a service-level agreement or maintenance subscription with verifiable fulfillment and periodic cancellation rights, rather than an unrestricted mechanism-design problem. The customer may leave *before implementation at every review*. Remaining contractual utility must exceed the publicly specified outside continuation $\underline{U}(h_t)$ at every positive-probability history h_t . The provider commits to its announced contingent protocol; the exit right does not let it retrospectively change prices or tiers. We consider contracts intended to retain the customer at all modeled reviews. A model in which termination is itself an optimized action would need an additional termination decision.

Stock can be contractually specified. It can also be identified from service and cost commitments: if s_z uniquely minimizes $C_z(S) - \xi f_z(S)$, $\xi > 0$, then $F^\pi \geq F^s$ and $C^\pi \leq C^s$ force $S_t = s_{z_t}$ at every reachable review. The nonnegative expected sum of the minimization gaps must be zero. EC Section EC.2 gives the boundary, vector, and relaxed-commitment statements. Thus the tier reduction need not assume away a stock decision that the accepted physical commitments already determine. Without such identification, stock must remain in the accepted optimization.

On a finite exogenous tree, let $w_n = \beta^{t(n)} \Pr(n) > 0$, $c_n = w_n a(z_n, s_{z_n})$, and $i \succeq n$ mean that i is in the subtree rooted at n . Multiplying conditional participation by w_n gives

$$\sum_{i \succeq n} c_i x_i \leq b_n, \quad b_n = \sum_{i \succeq n} w_i \nu f_{z_i}(s_{z_i}) - w_n \underline{U}(n). \quad (2)$$

This derivation imposes every continuation constraint directly. Ex ante participation alone is the special case retaining only the root row. It does not imply the nonroot inequalities. A fixed root payment, when stipulated in the agreement, gives $\sum_i c_i x_i = b_0$; it is a separate payment commitment, not an equality inferred from participation. General formulations below also allow a root inequality.

3.2 Optimized restrictions and switching

Write the accepted set, including the tier box, as

$$X_b = \{x \in C : Ax \leq b, Ex = e\}, \quad C = [\ell, u], \quad W_\lambda(x) = p^\top x - \frac{1}{2}x^\top Qx - \lambda T(x), \quad (3)$$

where $Q \succ 0$, $\lambda \geq 0$, and

$$T(x) = \sum_{a=1}^m \kappa_a |(Dx - h)_a|, \quad \kappa_a > 0.$$

The linear reward p includes net receipts and the linear part of convex operating costs, for example a quadratic cost around a service-specific efficient tier. It need not be proportional to the payment vector c . The general quadratic model and its synthetic examples are not asserted to be estimates of the stock institution above.

Rows of A include (2) and any other retained capacity constraints. E contains fixed payment or physical equalities. D can include temporal and cross-service differences; h incorporates an installed initial tier. The vector-service case uses the same stacked notation, and signed payment coefficients are allowed in (3). Only the specialized transfer lemma requires positive scalar coefficients. All accepted sets used below are nonempty and compact. Strong concavity makes each optimum unique.

A contractual restriction is a predeclared equality $Mx = m_0$, and

$$Y_b = X_b \cap \{Mx = m_0\}, \quad K_\lambda(b) = \max_{x \in Y_b} W_\lambda(x), \quad J_\lambda(b) = \max_{x \in X_b} W_\lambda(x), \quad \Delta_\lambda(b) = J_\lambda(b) - K_\lambda(b). \quad (4)$$

The matrices and commitments are the same on both sides. Pooling decisions at histories with the same permitted information gives one important M . The experiments also pool *amendments* $x - z^{\text{legacy}}$; then $m_0 = Mz^{\text{legacy}}$, which is generally different from pooling actual tiers. We keep that distinction explicit.

3.3 Continuation prices and relative transfers

For a comparator $z \in X_b$, put $g = p - Qz$. Convex optimality gives the following retained foundation.

Proposition 3.1 (Accepted balance). *The comparator z is optimal over X_b if and only if there exist $\mu \geq 0$, unrestricted η , $v \in N_C(z)$, and switching tensions s such that*

$$g = A^\top \mu + E^\top \eta + v + D^\top s, \quad \mu_i (A_i z - b_i) = 0, \quad (5)$$

with $s_a = \lambda \kappa_a \text{sign}(Dz - h)_a$ on nonzero differences and $|s_a| \leq \lambda \kappa_a$ on zero differences. If it is not optimal, a feasible sufficiently short direction has strictly positive directional gain. For optimality over Y_b , add $M^\top \xi$ to (5); ξ prices the restriction rather than a customer participation constraint.

Proof. The objective loss is the sum of a differentiable convex quadratic and a finite convex absolute-value function. Its subgradient plus the polyhedral normal cone characterizes a minimum. Polyhedral normals give the displayed multipliers, without a strict-feasibility assumption. The alternative follows from the directional derivative test for a convex function on a polyhedron. $\square$

In the scalar tree specialization, assume $c_n > 0$ and a fixed root payment. For any feasible z , define its continuation slack $\sigma_n = b_n - \sum_{i \geq n} c_i z_i$ at nonroot nodes. Let B have column n equal to -1 at row n and $c_n/c_{\text{pa}(n)}$ at its parent, with zeros elsewhere.

Lemma 3.2 (Benchmark-relative transfer coordinates). *The correspondence*

$$x = z + By, \quad y_n = -c_n^{-1} \sum_{i \geq n} c_i (x_i - z_i) \quad (6)$$

is a bijection between X_b and the relative transfers satisfying $y_n \geq -\sigma_n/c_n$, the original tier box, and every additional inequality or equality after substitution. The restriction $Mx = Mz$ becomes $MBy = 0$. Both directions of the map take linear work in the number of tree nodes.

Subtree telescoping proves the lemma; EC Section EC.3 gives the inverse explicitly. Negative transfers consume pre-existing continuation slack and cannot be discarded. The lemma is a representation, not a theorem about how much an optimized restriction costs. We next use the restriction prices and the slack in a value theorem.

4 Finite Continuation Prices on a Recombining Graph

We now give a constructive result, rather than recover a recursive certificate from an already solved scenario tree. Consider a finite acyclic public graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with one root, $N = |\mathcal{V}|$ reachable vertices, and $M = |\mathcal{E}|$ edges. A vertex includes the date, regime, and any predeclared public memory. At a nonterminal vertex, $p_{vw} > 0$ sums to one over its successors. The graph is an input description; its unfolding may have exponentially many histories. Discounting is $0 < \beta \leq 1$.

At an occurrence of vertex v , the provider chooses a continuous tier in $[l_v, h_v] \subseteq [0, 1]$, earns

$$g_v(x) = r_v x - \frac{1}{2} q_v x^2, \quad q_v > 0, \quad (7)$$

and incurs payment $a_v x$, where $a_v > 0$. Every occurrence must satisfy the conditional continuation cap B_v ; the root payment is exactly b_0 . These caps need not be slack. The specialization excludes intertemporal switching costs and other constraints coupling the current action to future actions except through the additive payment. It retains continuous decisions, heterogeneous rewards and payments, discounting, nonstationary Markov transitions, and binding participation at every review. The general switching model of Section 3 remains in the paper; Appendix C treats it with the inherited tier and remaining promise.

For a price $\eta \in \mathbb{R}$, let $H_v(\eta)$ be the optimal conditional reward minus η times the total conditional payment, subject to all caps in the subtree starting at v . Let $R_v(\eta)$ be the payment of that optimizer. A root equality price can be negative. The local tier response and the payment before imposing the current cap are

$$x_v(\eta) = \left[\frac{r_v - a_v \eta}{q_v} \right]_{l_v}^{h_v}, \quad d_v(\eta) = a_v x_v(\eta) + \beta \sum_w p_{vw} R_w(\eta), \quad (8)$$

where $[y]_l^h = \max(l, \min(h, y))$ and an empty successor sum is zero. The feasible payment interval is computed backward:

$$L_v = a_v l_v + \beta \sum_w p_{vw} L_w, \quad U_v = \min\{B_v, a_v h_v + \beta \sum_w p_{vw} U_w\}. \quad (9)$$

An infeasible successor, $L_v > U_v$, or $b_0 \notin [L_0, U_0]$ is reported as infeasibility. No grid projection changes the promise.

Theorem 4.1 (Finite-price construction). *Under the preceding assumptions, on every feasible subtree,*

$$R_v(\eta) = \min\{B_v, d_v(\eta)\}. \quad (10)$$

If $B_v \geq d_v(-\infty)$, set $\alpha_v = -\infty$. Otherwise, let α_v be the leftmost finite solution of $d_v(\alpha_v) = B_v$. The optimizer at incoming price η selects $x_v(s_v)$ and sends price s_v to every successor, where

$$s_v = \max(\eta, \alpha_v), \quad \chi_v = s_v - \eta \geq 0. \quad (11)$$

Each response is continuous, nonincreasing, and piecewise affine, with constant outer tails. Every response breakpoint belongs to

$$\mathcal{B} = \left\{ \frac{r_v - q_v h_v}{a_v}, \frac{r_v - q_v l_v}{a_v} : v \in \mathcal{V} \right\} \cup \{\alpha_v : \alpha_v > -\infty\}, \quad |\mathcal{B}| \leq 3N. \quad (12)$$

Consequently all responses can be constructed exactly from the graph in $O((N+M)N \log(N+1))$ arithmetic operations and comparisons, storing $O(N^2 + M)$ coefficient and graph entries. Coefficients are rational when the primitives are rational. These are arithmetic, not unit-cost bit-complexity, bounds.

Choose any finite η_0 with $R_0(\eta_0) = b_0$. The resulting policy is optimal for the exact root promise. Along a realized history its effective price is the running maximum of η_0 and the barriers encountered so far. Thus an optimal contract uses at most $N+1$ price labels and $N(N+1)$ public-vertex/incoming-price pairs. Neither the construction nor this policy graph requires an unfolded scenario tree or a full-tree primal–dual solution.

Proof. For a finite unfolding, the feasible tier set is compact and convex. Its quadratic reward is strictly concave on every nonfixed coordinate, so the optimizer is unique. All intermediate payments in (9) are attainable by convex mixing; induction proves the interval formula.

Assume the response assertion holds at the successors. Omit only the current cap. At a given η , the local tier and successor subproblems separate, giving (8). If its payment is at most B_v , this optimizer is feasible and no current multiplier is needed. Otherwise increase its price to $s = \alpha_v$ so that its payment is B_v . Such a finite crossing exists because $B_v \geq L_v$ and the lower tail is constant. The policy optimal for the problem without the current cap at price s is also optimal with that cap at price η : for every feasible policy with reward G and payment P ,

$$G - \eta P = (G - sP) + (s - \eta)P \leq H_v^{\text{uncapped}}(s) + (s - \eta)B_v.$$

The constructed policy attains equality. This proves both (10) and (11); it also proves complementarity $\chi_v(B_v - R_v(\eta)) = 0$. A plateau creates no difficulty: the stated leftmost crossing or any price on an equality plateau implements the same unique policy. If the entire response is constant, the cap is redundant whenever the subtree is feasible.

A local clipped affine payment has at most two breakpoints. Taking a weighted sum introduces no breakpoint outside the union of the summands' breakpoints. Truncating a continuous nonincreasing piecewise-affine function from above removes some breakpoints and introduces at most its one crossing with the cap. Induction in reverse topological order proves (12). This argument concerns the *union over graph vertices*, not a separate union over histories. At each vertex, merge sorted child and local breakpoint lists, add their affine coefficients, and locate the cap crossing by an interval scan. With at most $3N$ retained breakpoints per list, the displayed conservative arithmetic bound follows. Store the coefficient lists and graph adjacency; no ancestor-indexed price data are an input.

The continuous response takes every value in its feasible interval, including its constant tails. Root inversion therefore supplies a finite η_0 . For any policy with payment b_0 , optimizing reward minus η_0 times payment is equivalent to optimizing reward. The backward construction is hence optimal for the root equality. Repeated application of (11) gives the running maximum. Its incoming value belongs to $\{\eta_0\} \cup \{\alpha_v > -\infty\}$, proving the finite-label bound. Forward propagation on these pairs computes discounted occupation weights, rewards, and all conditional obligations without unfolding histories. $\square$

What makes this a compression theorem. The bound is relative to the supplied public graph, not to an unbounded family of history augmentations. Arbitrarily enlarging public memory can enlarge N ; the theorem instead constructs its own bounded price record on that fixed graph. The theorem generates its own thresholds before solving the root query. Each cap contributes at most one threshold because scalar prices are ordered and payment responses are monotone. The resulting finite union bounds both exact response storage and the history information retained by the optimizer. In contrast, the general shared-table representation in Appendix C.3 may reconstruct arbitrarily many supporting planes from extensive-form multipliers. We call that earlier statement *extensive-form-dual representability*; no discovery bound is attached to it. The finite-price theorem does not claim that Benders cuts are new, or that generic multistage convex programs enjoy this bound.

Corollary 4.2 (Several service coordinates under one commitment). *At vertex v , replace the scalar tier by d_v^{act} coordinates, a product interval, reward $\sum_i (r_{vi}x_i - q_{vi}x_i^2/2)$, and the scalar payment $\sum_i a_{vi}x_i$, with $q_{vi}, a_{vi} > 0$. Let $D = \sum_v d_v^{\text{act}}$. Theorem 4.1 holds with at most $2D + N$ global response breakpoints, at most $N + 1$ price labels, and $O(N(2D + N) + M)$ coefficient/graph storage. A weighted sum of the coordinate clipped responses replaces the first term in (8).*

Proof. The local Lagrangian separates by service coordinate and has at most $2d_v^{\text{act}}$ tier knots. The scalar cap still adds at most one threshold. Every remaining step of the preceding proof is unchanged. $\square$

Several *independent* commitments with separable feasible sets can be solved coordinate by coordinate. Coupled vector promises, dense intertemporal quadratic terms, or switching frictions do not inherit the or-

dered scalar reflection. Their retained promise-state formulation remains valid, but no finite-price complexity claim is made for them.

4.1 A compact exact optimality certificate

The construction also makes the optimality claim independently checkable. For a reached pair (v, η) , record its discounted occupation weight w , incoming and effective prices, action x , cap multiplier χ , and nonnegative lower/upper interval multipliers ℓ, u satisfying

$$r_v - q_v x - a_v s + \ell - u = 0, \quad \ell(x - l_v) = u(h_v - x) = \chi(B_v - R_v(\eta)) = 0. \quad (13)$$

Check the conditional payment identity and propagate weight $\beta w p_{vw}$ to each child with incoming price s . The scalar

$$\eta_0 b_0 + \sum_{(v, \eta)} w \left[\chi B_v + u h_v - \ell l_v + \frac{(r_v - a_v s + \ell - u)^2}{2q_v} \right] \quad (14)$$

is an extensive-form dual upper bound represented on the compact policy graph. Indeed, price propagation collects precisely the normalized ancestor cap multipliers on each payment coefficient; maximizing each unconstrained quadratic gives its squared term. With (13) and feasible promised payments, this bound equals the implemented reward. This certifies the exact *system-level* value; an error allowance is not multiplied by the number of replicated components.

5 How Much Contractual Memory Is Necessary?

The finite-price state yields an operational conclusion: the provider need not retain the complete sequence of customer reviews, but it may have to retain which past participation constraint has the highest effective payment price. Once thresholds and the query-specific root price are stored in a read-only table, the changing record needs at most $\lceil \log_2(N+1) \rceil$ bits. This counts the record index, not the storage or numerical precision of the threshold table. A new root promise can change the root price and requires a new root inversion.

A three-date example shows why the record cannot simply be dropped. The root tier is fixed at zero. Two equally likely intermediate regimes, A and B , both lead to the same terminal regime C . Payments and discounting are one, all nonfixed tiers lie in $[0, 1]$, and rewards are $-x^2/2$ at A, B and $2x - x^2/2$ at C . The intermediate caps are $1/4$ and 1 , the terminal cap is 1 , and the exact root promise is $1/2$. The full optimum has intermediate tiers zero and terminal tiers $1/4, 3/4$, with value $27/32$. Its effective terminal prices are $7/4, 5/4$. A date-regime table instead chooses the same terminal tier c . Its optimum is $c = 1/4$, with intermediate tiers $0, 1/2$, and value $13/32$. Thus retaining the price record creates the optimized gain $7/16$, not merely an improvement over an arbitrary fixed schedule. EC Section EC.1 gives the direct calculation.

The next result derives an entire value-memory frontier from contract primitives, rather than introduce a cost and differentiate a precomputed value function. The segmentation algorithm used to evaluate the frontier is standard; its interval cost and the sufficiency of ordered memory cells are the contractual conclusions.

Theorem 5.1 (Necessary labels and an exact memory-value frontier). *Fix distinct caps $0 < b_1 < \dots < b_k < 1$. Consider the preceding three-date architecture with k equally likely intermediate regimes, cap b_j in regime j , and one common terminal regime. Set the exact root promise to $k^{-1} \sum_j b_j$. The optimal full-information policy has intermediate tier zero and terminal tier b_j on branch j . Its value is*

$$J = \frac{1}{k} \sum_{j=1}^k (2b_j - b_j^2/2).$$

Any deterministic implementation attaining J needs at least k distinguishable memory symbols upon entering the terminal regime. Thus the linear worst-case order of the finite-price label bound cannot be improved to a constant independent of the public graph.

If at most m terminal memory symbols are available, an optimal encoding partitions the ordered caps into at most m contiguous nonempty cells. For a cell $i, \dots, j$, the optimal common terminal tier is b_i , and its contribution to the loss relative to J is

$$C(i, j) = \frac{2 - b_i}{k} \sum_{h=i}^j (b_h - b_i). \quad (15)$$

The exact minimum loss for m symbols is $D(m, k)$, computed from

$$D(s, j) = \min_{s-1 \leq i < j} \{D(s-1, i) + C(i+1, j)\}, \quad D(0, 0) = 0, \quad (16)$$

with infeasible indices assigned $+\infty$. Prefix sums give an $O(mk^2)$ arithmetic algorithm and $O(mk)$ storage. A device with B writable bits therefore attains value $J - D(\min\{2^B, k\}, k)$ under this institution.

Proof. The root promise is the mean of the conditional caps. Every branch cap must consequently bind: a strict inequality on any branch would make the root promise too small. If branch j has terminal tier c_j , its intermediate tier is therefore $b_j - c_j$, with $0 \leq c_j \leq b_j$. Its reward is

$$f_j(c_j) = 2c_j - c_j^2/2 - (b_j - c_j)^2/2, \quad f'_j(c_j) = 2 + b_j - 2c_j > 0.$$

Thus full information selects $c_j = b_j$. Strict concavity and positive branch probabilities imply a unique optimal tier vector. Since the terminal public regime is common and the b_j are distinct, two branches sharing a terminal memory symbol cannot both implement their required optimal terminal tiers. At least k symbols are necessary, and storing the branch's reflected price is sufficient. This construction has $N = k + 2$ public vertices.

For any memory cell, the common terminal tier cannot exceed its smallest cap. Every f_j is increasing on the feasible interval, so that minimum cap is optimal. Subtracting $f_j(c)$ from $f_j(b_j)$ gives $(b_j - c)(2 - c)$, proving (15) for an ordered cell. Now fix the set of cell minima. Assigning branch j to any eligible minimum $c \leq b_j$ has loss $(b_j - c)(2 - c)$, whose derivative with respect to c is $-2 - b_j + 2c < 0$. Each branch therefore belongs to the largest selected minimum not exceeding its cap. This reassignment preserves each selected minimum and makes every cell contiguous, without increasing loss. Dynamic programming over the final

cell proves (16). Additional symbols can split cells and never reduce value; hence exactly $\min(m, k)$ symbols are sufficient for an optimum with at most m . The bit statement follows by counting available symbols. $\square$

For $k = 8$ and $b_j = j/9$, one symbol loses $119/162$ relative to the full optimum; two symbols lose $5/18$, and three lose $49/324$. The complete frontier and its minimizing partitions are retained in the data. A capacity of two bits permits four symbols, not three. The theorem prices the effect of a concrete memory technology; a declared installation or maintenance cost can subsequently be compared with these gains. It neither estimates that cost nor presumes a calibrated application.

6 Optimizing a Shared Information Restriction

The finite-price construction also changes how an optimized comparator is obtained. There are two distinct operations: choosing one table before uncertainty, and executing accepted recourse around that table. They must not be interchanged.

6.1 An exact graph-sized formulation at zero release

Let $c(v) \in \{1, \dots, K\}$ be the permitted information cell and require every occurrence of v to use tier $u_{c(v)}$. A time-only table has one coordinate per date; a public-regime table can have one per vertex. Let π_v be the unconditional reach probability of v , obtained by forward propagation, and let $w_v = \beta^{t(v)} \pi_v$. A fixed table's conditional remaining payment, denoted y_v , satisfies

$$y_v = a_v u_{c(v)} + \beta \sum_w p_{vw} y_w, \quad y_v \leq B_v, \quad y_0 = b_0. \quad (17)$$

This is a system of N payment equations and N caps, not one row for every history.

Proposition 6.1 (Optimized shared-table condensation). *For the public graph institution, the optimized zero-release comparator equals the convex quadratic program*

$$K = \max_{u, y} \left\{ \sum_v w_v g_v(u_{c(v)}) : (17), l_v \leq u_{c(v)} \leq h_v \right\}. \quad (18)$$

Its extended formulation has $K + N$ variables and $O(N + M + K)$ nonzero coefficient occurrences, after intersecting tier intervals within each information cell. With incoming absolute-value switching costs, exact condensation still holds: add the root switching cost and, for each edge (v, w) , its discounted reach-weighted cost of $|u_{c(w)} - u_{c(v)}|$. Introducing one epigraph variable per edge preserves graph-sized convex representability.

Proof. Under a fixed table, every future action and primitive depends only on the displayed public vertex. Conditional expectation and backward induction therefore give the same remaining payment at every occurrence of that vertex, namely (17). Every feasible table in the history formulation satisfies these equations and all displayed caps. Conversely, a feasible solution implements the same table at every history and hence

satisfies every continuation inequality. Summing the common local rewards with the forward reach weights gives (18). Switching on an edge uses the two table entries, so its expected cost is also obtained by edge occupation weights. There is no omitted inherited action when the whole table is fixed. The extra root term uses the stipulated initial tier. $\square$

The proposition is not a new quadratic-programming algorithm. It establishes when the *optimized* contractual restriction can be compiled directly on the public input graph. Combining it with Theorem 4.1 yields both sides of the adaptation premium without full-history enumeration in the no-switching subclass. The public-only table remains a restriction: Theorem 5.1 shows why the full optimum need not be attainable with it. The graph-sized formulation and the finite-price controller thus solve different policy classes on the same primitives.

6.2 An exact fixed-table oracle at positive release

Fix one global table and permit $|x_h - u_{c(h)}| \leq \delta \omega_{c(h)}$. For fixed (u, δ) , intersect each corridor with its physical interval and apply Theorem 4.1. This constructs an exact value and a feasible policy, or detects an empty root payment interval. The physical primitives, graph, and root promise remain fixed across table queries.

Proposition 6.2 (Tree-free table supergradients). *At a feasible fixed-table query, the finite-price construction supplies an exact value $V(u, \delta)$ and corridor multipliers $\rho_{v,\eta}^+, \rho_{v,\eta}^- \geq 0$. Aggregate them with the compact pair occupation weights:*

$$G_c = \sum_{(v,\eta):c(v)=c} w_{v,\eta}(\rho_{v,\eta}^+ - \rho_{v,\eta}^-), \quad H = \sum_{(v,\eta)} w_{v,\eta} \omega_{c(v)}(\rho_{v,\eta}^+ + \rho_{v,\eta}^-). \quad (19)$$

For every other feasible table/release pair $(\tilde{u}, \tilde{\delta})$,

$$V(\tilde{u}, \tilde{\delta}) \leq V(u, \delta) + G^\top (\tilde{u} - u) + H(\tilde{\delta} - \delta). \quad (20)$$

Thus the fixed-table oracle constructs exact supporting cuts for a Benders-style shared-design master without starting from an extensive-form optimum.

Proof. The local stationarity residual in (13) assigns a nonnegative upper multiplier at an active upper endpoint and a nonnegative lower multiplier at an active lower endpoint. Assign that multiplier to an active physical bound or to the corridor attaining that endpoint; either allocation is valid at a tie. At a zero-width intersection, split according to the sign of the residual. Together with the reflected cap prices and root equality price, these are full Lagrange multipliers for the fixed-table optimum. The compact weights aggregate identical public-history occurrences of this certificate.

Keep these multipliers fixed and change only the corridor right-hand sides. Their linear change in the dual objective is exactly the right-hand side of (19). Weak duality gives (20); complementarity at the anchor makes its constant exact. Although the old price labels organize this stored certificate, its weak-duality bound applies to every new feasible policy, not only policies following those labels. $\square$

A finite collection of these cuts produces the usual shared-design upper master, and feasible oracle policies provide lower bounds. Optimizing a *continuous* shared table with positive release is not thereby assigned a finite exact cut count. The graph-sized exact comparator applies at zero release; the polynomial finite-price bound applies to each fixed-table oracle. Distinguishing these results prevents an existence statement about supporting planes from becoming an unsupported complexity claim about the outer design problem. The inherited implementation-cost analysis, release rents, and switching results remain in Appendices A–D.

7 Exact Computational Evidence

The new experiment operates on one coupled, recombining stochastic graph per instance. A root has two or three possible successors, and every subsequent public regime can transition to every next-period regime. Rewards, curvatures, payment coefficients, and continuation caps vary with date and regime. The exact root promise couples decisions across all reachable successors. The experiment does not increase dimension by replicating independent two-period blocks.

The main response suite contains 35 instances, including 64 decision dates, fixed shared-table corridors, root promises on both boundary tails, a negative equality price, tied barriers, fixed actions, and a cap binding at its minimum feasible payment. The independent verifier checks entire piecewise-affine response identities, all reached state flows and local KKT conditions, conditional participation, root payment, and equality of the implemented reward with its dual upper bound. It performs 147,157 exact assertion groups. Two deliberately infeasible inputs are rejected. The full certificate files, not only summary statistics, are retained.

Table 1 reports the main scaling observations. The 64-date, three-regime instance has 190 public vertices, 332 distinct response knots, and 580 reached price pairs. Its ternary unfolding would contain $(3^{64} - 1)/2$ decision histories. No such tree is constructed by the production solver. The response construction and independent audit took approximately 1.19 and 3.51 seconds in the recorded run; these are single-run Python rational-arithmetic observations, not a speedup estimate or a claim about a tuned general-purpose solver. Coefficient bit lengths and serialized certificate sizes are reported alongside counts. The numerical evidence supplements the proved graph-size bound; it is not used to infer worst-case complexity.

Fourteen small instances are separately unfolded. Their conditional-incidence matrices satisfy exact rational primal feasibility, stationarity, and complementarity. A feasibility linear program followed by a floating-point convex quadratic solve produces a largest absolute value discrepancy of approximately 2.33×10^{-12} . The numerical solver is only a cross-check: its termination flag is not the certificate. Eleven optimized time-only table comparators are solved and verified by exact rational KKT conditions on the independently reconstructed graph-sized quadratic programs. Their nonnegative adaptation premiums compare two optimized classes on identical primitives.

A further 54 perturbed table/release queries satisfy the exact supporting-cut inequalities, and a two-service instance checks the scalar-commitment extension. Eight memory families reach 32 distinct renewal regimes; all 277 set partitions for sizes two through six are enumerated independently to check the ordered-cell frontier. Table 2 gives the eight-regime frontier. No implementation costs are fitted to make a chosen memory architecture appear optimal.

The inherited evidence is retained as a separate, fully typeset computational appendix, with its original source, raw records, and interpretation. In particular, the 83/84 cache-refresh result and the absence of a measured cache speed advantage remain adverse findings. The prior learned routes also retain their unfavorable matched-quality results. Those observations concern different algorithms and designs; they are neither erased nor relabeled as measurements of finite-price computation. EC Section EC.1 specifies the new design and verification boundaries.

8 Conclusions

Nested continuation participation has two distinct consequences. For general accepted adaptation it creates scarcity prices, release rents, and a need to carry outstanding promises. For the separable no-switching subclass, those prices admit a stronger construction: each public cap contributes at most one threshold, and an optimizer carries a finite running-maximum record rather than the history tree. The resulting exact graph algorithm, its memory lower bound, and the memory-value frontier give substantive content to the information architecture.

The shared table remains a commitment chosen once. Its zero-release value has a graph-sized formulation, and its positive-release subproblems have exact tree-free price oracles under the finite-price assumptions. Broader switching and correlated-certificate results remain valid under their separately stated assumptions. They are not used to infer a complexity bound that their proofs do not supply. The computational record distinguishes exact structural checks, single-run cost measurements, and the unchanged adverse historical evidence.

Data and Code Accessibility

The code/data package contains all rational primitive graphs, compact primal–dual certificates, independent verifiers, small-tree matrix and numerical comparisons, optimized-table KKT records, memory partitions, scripts, and build checks. The complete R28 manuscript, companion, and reader files are preserved byte for byte, as are all earlier repository derivations. Its complete numerical narrative and tables are additionally typeset in the retained computational appendix. No historical mathematical statement, proof, or unfavorable result is silently discarded. No field data are used. Repository delivery is for author/referee scrutiny, not a journal submission or a representation of editorial acceptance.

Print Appendices: General Accepted Adaptation

The following results preserve the broader mathematical framework. The finite-price theorem does not require the deployment certificates or conditional continuous-state approximation rates. These appendices state what remains valid when the tractable no-switching specialization is not imposed; the electronic companion retains their detailed proofs and all qualifications.

A The Value Frontier of Releasing a Restriction

Fix b, λ and the optimized comparator $z = \arg \max_{Y_b} W_\lambda$. Let $\omega \in \mathbb{R}_{++}^k$ set the units of permitted deviations from the restriction. Define

$$V(t) = \max\{W_\lambda(x) : x \in X_b, |Mx - m_0| \leq t\omega\}, \quad t \geq 0. \quad (21)$$

Thus $V(0) = K_\lambda(b)$. The release does not relax participation: it permits previously pooled decisions to differ while preserving every accepted constraint. The full value is reached for finite t , because X_b is compact. Scaling a row of M requires the same scaling of its ω ; otherwise it changes the economic release unit.

A.1 An exact release segment and its first bottleneck

To state the explicit segment, fix a regular active cell at z . Collect permanent equalities, active accepted inequalities and tier bounds, and zero switching differences into $Rx = r$. Retain the signs of all nonzero switching differences and absorb their linear cost into $\bar{p}$. Let R and $[R^\top, M^\top]^\top$ have full row rank. Suppose the active accepted-inequality multipliers are strictly positive, zero-edge tensions are strictly within their capacities, inactive constraints and nonzero switching signs have positive clearance, and the restriction multiplier ξ has no zero coordinate. These are local regularity conditions, not assumptions of the global frontier in the theorem. Rows with zero switching coefficient need not be included as kink constraints.

Choose any matrix Z whose columns form a basis for $\ker R$, and set

$$H = Z^\top QZ, \quad S = [MZH^{-1}Z^\top M^\top]^{-1}, \quad P = ZH^{-1}Z^\top M^\top S, \quad d = Pv, \quad v_j = \omega_j \text{sign } \xi_j. \quad (22)$$

These objects are independent of the choice of basis. S is the marginal curvature of releasing the restriction after the currently binding accepted constraints have been enforced.

Theorem A.1 (Restriction-release frontier). *For (3)–(21), V is continuous, nondecreasing, concave, and piecewise quadratic. Its unique tier policy is continuous and piecewise affine. At almost every $t > 0$,*

$$V'(t) = \omega^\top (\xi^+(t) + \xi^-(t)), \quad V(t) - V(0) = \int_0^t \omega^\top (\xi^+(s) + \xi^-(s)) ds, \quad (23)$$

where $\xi^\pm$ are optimal multipliers on the two release bounds. At differentiability points the displayed directional price is independent of the choice of optimal multipliers.

Under the regular-cell conditions, let t_1 be the first nonnegative time at which an inactive accepted constraint or tier bound binds, a nonzero switching difference reaches zero, an old active-inequality multiplier reaches zero, a zero-edge tension reaches a capacity, or a component of $\xi - tSv$ reaches zero in its original signed direction. Then $t_1 > 0$ and, for $0 \leq t \leq t_1$,

$$x(t) = z + td, \quad V(t) - K_\lambda(b) = t \sum_j \omega_j |\xi_j| - \frac{1}{2} t^2 v^\top S v. \quad (24)$$

Each event time is an explicit linear ratio obtained from the primal and dual paths. In particular, a currently

slack continuation row can be the first bottleneck only at

$$t_n = \frac{\sigma_n}{\sum_{i \geq n} c_i d_i} \quad \text{when } \sum_{i \geq n} c_i d_i > 0; \quad (25)$$

otherwise that row consumes no slack along this segment. In transfer coordinates this is exactly the time at which its lower bound $y_n \geq -\sigma_n/c_n$ is reached.

Proof. Mixing policies feasible at two release levels proves concavity; nesting proves monotonicity. Introduce epigraphs for the absolute values. For fixed active inequalities and switching signs the KKT equations are linear in the parameter, so the unique tier solution is affine and its value is quadratic on that cell. There are finitely many active patterns. Dependent equations may be removed and a finite refinement made; no nondegeneracy assumption is needed for this global piecewise description. Compactness and uniqueness give continuity. Optimal multipliers support the concave value and give (23); integration is valid for this continuous piecewise-quadratic function.

For the explicit segment, a prescribed restriction displacement δ has the cell optimum $x = z + P\delta$. Solving the equality-constrained quadratic and using $RP = 0$, $MP = I$, and $P^\top QP = S$ gives the gain $\xi^\top \delta - \delta^\top S\delta/2$. Its maximum over $|\delta| \leq t\omega$ is at $\delta = tv$ while $\xi - tSv$ retains its signed direction. The old active multipliers and kink tensions also evolve affinely. Until one of the listed events, their signs and capacities, the remaining primal inequalities, and the nonzero switching signs all remain valid. They therefore supply the *full* KKT certificate, not just a solution restricted to the cell. Continuity includes the first event. Substitution into a subtree inequality gives (25). The detailed dual path and degeneracy qualifications are in EC Section EC.4. $\square$

Write $a = \sum_j \omega_j |\xi_j|$ and $q = v^\top Sv > 0$. For a desired gain $g \in [0, V(t_1) - K]$, the smallest release on this initial segment is

$$t(g) = \frac{a - \sqrt{a^2 - 2qg}}{q}. \quad (26)$$

No accepted improvement can attain this gain with a smaller release, because (24) is the optimized value, not the gain of a selected direction. At later events one changes the active cell and applies the same equality-constrained calculation. The number of cells need not be polynomial; the theorem is an exact structural path, not a universal complexity claim. If a regularity condition fails, (23) and the global frontier remain valid, but the signed-corner formula is replaced by the appropriate small release quadratic program or a degenerate active-set refinement.

A.2 A complete three-node frontier

Consider a root and two children, $c_i = 1$, tiers in $[0, 1]$, root payment $x_0 + x_1 + x_2 = 3/2$, and child caps $x_1, x_2 \leq 3/4$. Let

$$W(x) = x_1 - \frac{1}{2}(x_0^2 + x_1^2 + x_2^2), \quad Y = \{x \in X : x_1 = x_2\}.$$

The optimized restriction is $z = (1/6, 2/3, 2/3)$ with value $5/24$. Releasing $|x_1 - x_2| \leq t$ gives

$$V(t) - K = \begin{cases} t/2 - t^2/4, & 0 \leq t \leq 1/6, \\ -1/48 + 3t/4 - t^2, & 1/6 \leq t \leq 3/8, \\ 23/192, & t \geq 3/8. \end{cases} \quad (27)$$

The first policy segment is $(1/6, 2/3 + t/2, 2/3 - t/2)$. Child 1 has slack $1/12$, consumes it at rate $1/2$, and therefore binds at $t = 1/6$. The second segment is $(t, 3/4, 3/4 - t)$. At $t = 3/8$, the release price becomes zero and the full optimum $(3/8, 3/4, 3/8)$ is reached. Its relative transfer is $(-1/12, 7/24)$; the negative coordinate uses exactly the initially available child-1 slack. Thus the continuation slack determines an actual change in the optimal policy path, rather than only the domain of a coordinate map. EC Section EC.5 supplies exact primal and dual witnesses for both segments.

B Capacity, Friction, and Optimized Comparator Rents

B.1 Relaxing a continuation can reduce the adaptation premium

Let $b(t) = b^0 + td$, with $d \geq 0$, and suppose $Y_{b(t)}$ is nonempty on the interval considered. Both optima in (4) are reoptimized for each t .

Theorem B.1 (Comparator-adjusted continuation rents). *The full and restricted values are continuous, nondecreasing, concave, and piecewise quadratic along the budget path. If $\mu_X(t)$ and $\mu_Y(t)$ are optimal accepted-inequality multipliers, then, at almost every t ,*

$$\frac{d}{dt} \Delta_\lambda(b(t)) = d^\top (\mu_X(t) - \mu_Y(t)), \quad \Delta_\lambda(b(v)) - \Delta_\lambda(b(u)) = \int_u^v d^\top (\mu_X - \mu_Y) dt. \quad (28)$$

Consequently, the adaptation premium is nondecreasing along the path if and only if the rent difference is nonnegative almost everywhere. At a degenerate point, the right directional value derivative along d is the minimum of $d^\top \mu$ over that problem's optimal multiplier set; one must compute the two directional derivatives separately, rather than subtract arbitrary selected multipliers.

Proof. Mixing and nesting give the first two value properties. Fixed-matrix parametric quadratic programming gives the piecewise description as in Theorem A.1. Weak duality makes each accepted multiplier a supporting price for the budget. At differentiability points all such directional prices coincide with the derivative. Subtract and integrate the two value derivatives. A continuous piecewise-quadratic function is nondecreasing precisely when its derivative is nonnegative almost everywhere. The one-sided rule follows by taking the directional derivative of the infimum of affine-in-budget dual objectives; EC Section EC.6 supplies the polyhedral qualification. $\square$

An exact example makes the distinction material. Keep the preceding three-node objective and root payment, but set $x_1 \leq b$, $x_2 \leq 3/4$, with $b \in [1/4, 2/3]$. The time-only restriction remains $x_1 = x_2$. The full

and restricted optima are

$$x^X(b) = ((3/2 - b)/2, b, (3/2 - b)/2), \quad x^Y(b) = (3/2 - 2b, b, b),$$

with cap prices $\mu_X = 7/4 - 3b/2$ and $\mu_Y = 4 - 6b$. Both values increase, but

$$\Delta(b) = \frac{9}{16}(2b - 1)^2, \quad \Delta'(b) = \frac{9}{4}(2b - 1). \quad (29)$$

The premium falls to zero as b rises to $1/2$, then rises again. A claim that more participation slack necessarily makes adaptation more valuable would get the sign wrong on half this interval. The relevant scarcity measure is the difference of the optimized contractual rents.

B.2 Friction thresholds when the restriction is reoptimized

For fixed X, Y , let x_λ and z_λ be their optima. A fixed-comparator balance test is not enough when z_λ changes with friction.

Proposition B.2 (Optimized-comparator friction path). *The adaptation premium is continuously differentiable for $\lambda > 0$, with a right derivative at zero, and*

$$\Delta'(\lambda) = T(z_\lambda) - T(x_\lambda). \quad (30)$$

On every bounded friction interval its zero set is a finite union of closed intervals and isolated points. It can be computed from the restricted parametric policy path without solving a full-class optimization at every friction: refine that path at changes of its accepted active rows and switching signs, and on each resulting cell impose (5) at the affine z_λ . This is a linear feasibility problem jointly in λ and the full-class prices; its projection on λ is an interval, possibly empty or a point. The union of these projections is exactly the zero set. This is a structural reduction, not a general complexity improvement. There is no general upward-ray conclusion for arbitrary restrictions.

Proof. Uniqueness and compactness imply continuous optimizers. The envelope argument gives $J'(\lambda) = -T(x_\lambda)$ and $K'(\lambda) = -T(z_\lambda)$, including active-set changes. Both paths are piecewise affine. On a refined cell, every sign in (5) is fixed or zero, z_λ is affine, and tension bounds and stationarity are linear in the displayed variables. The full balance test is necessary and sufficient. A projection of a polyhedron onto a line is an interval. Finitely many cells establish the assertion; the method does not bound their number polynomially. $\square$

For example, use the root payment $3/2$, the box and child caps above, $p = (1/2, 3/4, 1/4)$, $Q = I$, and $T(x) = |x_1 - x_0| + |x_2 - x_0|$. A restriction $x_1 - x_2 = 1/2$ has optimized policy $z = p$ for every λ : its switching cost is already the smallest possible under that restriction. The full optimum is $(1/2, 3/4 - \lambda, 1/4 + \lambda)$ until

$\lambda = 1/4$, and is constant thereafter. Therefore

$$\Delta(\lambda) = \begin{cases} \lambda^2, & 0 \leq \lambda \leq 1/4, \\ \lambda/2 - 1/16, & \lambda \geq 1/4. \end{cases} \quad (31)$$

More friction increases the value of removing a restriction that forces switching. This does not contradict constant-comparator critical-friction results: their restriction and zero-switching benchmark are different.

B.3 Which information expansion creates the value?

Let $Y_0 \subseteq \dots \subseteq Y_k = X$ be nonempty predeclared classes and $K_j = \max_{Y_j} W$. Then

$$J - K_0 = \sum_{j=1}^k (K_j - K_{j-1}), \quad K_j - K_{j-1} \geq 0. \quad (32)$$

The identity is elementary; its economic use requires each K_j to be optimized under the *same* accepted constraints. A release path between adjacent equality-defined classes prices that particular increment through Theorem A.1. The ordering matters when information features interact; the increments are not automatically Shapley values or unique causal attributions.

The fixed class describes an unchanging amendment or tier. The time-only class describes a preannounced schedule. The current-regime class requires public regime measurement and a date–regime decision table. A two-period-memory class additionally requires retaining the preceding regime. The full-history class has no such information pooling. These are operational restrictions, not an assertion that time-only is the universal status quo. Current regime alone can be insufficient even when exogenous primitives are Markov, because the inherited tier and the outstanding promise are endogenous state variables. The next section establishes the corresponding sufficient-state formulation.

C Promised-Payment Recursion and Horizon Complexity

Suppose the regime has a finite public Markov transition matrix P_t , the one-period net payment is $a_t(z)x$, and the reward is $g_t(z, q, x)$. Suppose further that each conditional payment cap $\bar{b}_t(z)$ and the feasible actions depend only on the displayed state and date. A general history-dependent outside option does not satisfy this assumption merely because demand is Markov.

Let $\mathcal{V}_t(z, q, b)$ denote the maximum conditional reward from date t with an *exact* remaining discounted payment promise b . Put $\mathcal{V}_T(z, q, 0) = 0$ and $\mathcal{V}_T(z, q, b) = -\infty$ for $b \neq 0$. For $b \leq \bar{b}_t(z)$, define

$$\begin{aligned} \mathcal{V}_t(z, q, b) &= \max_{x, \{b_{z'}\}} g_t(z, q, x) + \beta \sum_{z'} P_t(z, z') \mathcal{V}_{t+1}(z', x, b_{z'}) \\ \text{subject to} \quad & a_t(z)x + \beta \sum_{z'} P_t(z, z') b_{z'} = b, \\ & b_{z'} \leq \bar{b}_{t+1}(z') \quad \text{for positive-probability children.} \end{aligned} \quad (33)$$

Outside this cap, set the value to $-\infty$. A zero-probability successor imposes no obligation. For a root upper budget rather than an equality, maximize the root value also over permitted b .

Theorem C.1 (Exact sufficient-state implementation). *Under the stated Markov sufficiency and compact-attainment conditions, (33) has the same value as the full accepted history-tree problem with its specified root promise. There exists an optimal deterministic contract implemented using only (t, z, q, b) . Conversely, a feasible recursion policy induces all tree continuation constraints. With several additive commitments, the result holds with a vector of remaining commitments. A history-dependent cap, nonadditive global resource, or omitted physical state must be included in the state or treated on the full tree; otherwise the claimed equivalence need not hold.*

Proof. For any history policy, take its conditional remaining payments in each child subtree as the child promises. They satisfy the equality and all child caps, and induction bounds its conditional reward by the right side. Conversely, combine an attained current maximizer with child maximizers at the allocated promises. Induction gives every child obligation, while the displayed equality gives the parent’s exact promise. This proves equality and constructs a state-based policy. The vector case applies the same induction to each additive commitment. $\square$

Conditional normalization is essential. Multiplying the conditional budget at node n by w_n recovers (2). Within (33), the child weights are $\beta P_t(z, z')$, not unconditional probabilities multiplied a second time. Promises are decision variables allocated across next-period regimes; they are not forecasts fixed independently of the current decision.

A branching factor B and T decision dates can create $N = (B^T - 1)/(B - 1)$ history nodes. The linear transfer map is linear in N , not in T . The recursion avoids materializing those histories when the displayed state is sufficient, but its promise state is continuous in the continuous-tier model. A grid with G promise points, Q_0 inherited-tier points, S_0 regimes, and k commitment coordinates has order $TS_0Q_0G^k$ value entries; choosing child allocations is an additional cost and can grow exponentially in the number of successors. Such a grid is an approximation unless the contract institution itself has the stated finite actions and attainable promises. Our exact recursion checks use the latter finite-action interpretation, and exhaustively compare every attainable root promise with the full history enumeration. No interpolation theorem or unrestricted polynomial-horizon guarantee is inferred from them.

C.1 Continuous promises with executable error certificates

The sufficient-state statement is not yet an approximation guarantee. Interpolating a value table can violate an exact promise or admit a customer continuation that the contract excludes. We next construct both a feasible policy and a value bound without rounding the promise to a grid point.

Specialize to $x, q \in [0, 1]$, $a_t(z) > 0$, and

$$g_t(z, q, x) = r_t(z)x - \frac{1}{2}d_t(z)x^2 - \ell_t(z)|x - q|, \quad d_t(z) > 0, \quad \ell_t(z) \geq 0. \quad (34)$$

Only the Markov continuation caps and the exact payment commitment are imposed in this specialization;

additional nonlocal constraints require additional state. Write $I_t(z) = [L_t(z), U_t(z)]$ for feasible remaining promises. With $L_T = U_T = 0$, their endpoints satisfy

$$L_t(z) = \beta \sum_j P_t(z, j) L_{t+1}(j), \quad U_t(z) = \min \left\{ \bar{b}_t(z), a_t(z) + \beta \sum_j P_t(z, j) U_{t+1}(j) \right\}. \quad (35)$$

A state is infeasible if a positive-probability successor is infeasible or the displayed lower endpoint exceeds the upper endpoint. All statements below are on feasible states; $\mathcal{D}_t(z) = [0, 1] \times I_t(z)$ includes the boundary.

At each date and regime, retain finitely many anchors (q_i, b_i, l_i) and affine upper planes $H_k(q, b) = A_k q + B_k b + C_k$. An anchor includes a feasible current action and mixtures of child anchors with the exact inherited tier and remaining promise, not just a scalar value. The anchor coordinates span $\mathcal{D}_t(z)$. Define

$$\begin{aligned} \underline{V}_t(z, q, b) &= \max_{\alpha \geq 0} \left\{ \sum_i \alpha_i l_i : \sum_i \alpha_i = 1, \sum_i \alpha_i (q_i, b_i) = (q, b) \right\}, \\ \bar{V}_t(z, q, b) &= \min_k H_k(q, b). \end{aligned} \quad (36)$$

To generate a parent plane, choose any convex mixture of child planes for each successor j , with coefficients (A_j, B_j, C_j) . Suppress the parent date and regime in a, r, d, ℓ, P . For any $s \in [-\ell, \ell]$ and $\eta \in \mathbb{R}$, put

$$\begin{aligned} H(q, b) &= sq + \eta b + C, \\ C &= \psi_d \left(r - s - a\eta + \beta \sum_j P_j A_j \right) + \beta \sum_j P_j \left[C_j + \max_{v \in I_{t+1}(j)} (B_j - \eta)v \right], \\ \psi_d(v) &= \max_{0 \leq x \leq 1} \{vx - dx^2/2\}. \end{aligned} \quad (37)$$

The maximizer in ψ_d is $\min\{1, \max\{0, v/d\}\}$. Thus a proposed plane has a checkable global constant even when its prices were obtained only approximately.

Theorem C.2 (Promise-preserving continuous-state envelopes). *Under (34) and the stated Markov assumptions, the interval recursion (35) is exact and $\mathcal{V}_t(z, \cdot, \cdot)$ is concave on $\mathcal{D}_t(z)$. Start with zero terminal values and construct anchors and planes backward as above. If every anchor reward is a lower bound for its displayed feasible action and child mixtures, and every plane constant is at least (37), then*

$$\underline{V}_t(z, q, b) \leq \mathcal{V}_t(z, q, b) \leq \bar{V}_t(z, q, b) \quad \text{for every } (q, b) \in \mathcal{D}_t(z). \quad (38)$$

At a queried state, an optimizing mixture in (36), followed by its mixed action and child promises, gives a deterministic accepted policy with value at least $\underline{V}_t$. Repeating this procedure using only the current augmented state preserves every continuation cap and the exact root payment. Its loss is at most $\bar{V}_t - \underline{V}_t$.

A uniform certificate requires no unproved smoothness rate. Triangulate the anchor rectangle using stored grid anchors. For a triangle C_0 with vertices i , let

$$\varepsilon_t(z) = \max_{C_0} \min_k \max_{i \in \text{vert}(C_0)} \{H_k(q_i, b_i) - l_i\}. \quad (39)$$

Then $0 \leq \mathcal{V}_t - \underline{V}_t \leq \varepsilon_t(z)$ throughout the rectangle. The order of the minimum and maximum in (39) matters: checking only the minimum-plane gap at each vertex does not in general certify the interior.

Proof. The attainable parent promises are the sum of the action-payment interval and the discounted child intervals, intersected with the parent’s cap. This proves (35) by induction. Deterministic convex combinations of two feasible subtree policies preserve the affine payment equalities and all continuation inequalities. Joint concavity of (34), including its inherited-tier argument, proves value concavity.

For a lower anchor, combine the displayed action with the child mixtures. Induction gives the stated reward lower bound. A convex mixture of parent witnesses remains feasible at its mixed inherited tier and promise; concavity bounds its reward from below by the mixed anchor rewards. At the next review, reoptimizing the child mixture cannot reduce that lower bound. This proves the executable lower envelope without storing a separate policy for every history.

For the upper bound, replace each child value by its chosen affine majorant, use $-\ell|x - q| \leq -s(x - q)$, and subtract η times the promise equation. Maximizing the separated action and child-promise terms gives (37). This is weak duality and requires neither an interior promise nor exact optimizer multipliers. The minimum of the resulting planes remains an upper bound.

On a triangle, barycentric interpolation of its anchor rewards is no larger than $\underline{V}_t$. For any one plane, the difference from that interpolation is affine and is maximized at a vertex. Minimize this bound over planes, then maximize over triangles. This proves (39). $\square$

The construction compresses *certificates* as well as values. With at most A_0 anchors and H_0 planes per date–regime and at most B_0 successors, dense witness storage is of order $TS_0B_0(A_0^2 + H_0^2)$, not the number of histories. Sparse mixtures reduce storage further. Evaluating a policy still requires a small mixture linear program; building each table requires optimization over child allocations. Neither the anchor count needed for a prescribed tolerance nor the size of rational arithmetic is bounded independently of the problem data. Concave interpolation and stochastic dual cuts are established ideas (Pereira and Pinto, 1991); the contractual conclusion here is that continuous promise interpolation can be accompanied by exact participation feasibility, recursive price witnesses, and an a posteriori policy-loss bound. No such conclusion is inferred from finite-action enumeration alone. In particular, deterministic mixing is invalid for indivisible tiers unless mixtures themselves are admissible actions.

C.2 What is lost by omitting part of the augmented state?

Theorem C.1 gives zero value to additional history once the displayed state is sufficient. A smaller decision table can still lose value. That loss can be bounded from a certified full-class decision rather than attributed vaguely to “adaptivity.”

Let $\mathcal{A}$ be the affine subspace imposing a predeclared pooling of histories, and let $Y = X \cap \mathcal{A}$ be nonempty. For the general objective in Section 3, define $\text{dist}_Q(x, \mathcal{A}) = \inf_{y \in \mathcal{A}} \|x - y\|_Q$, where $\|u\|_Q^2 = u^\top Q u$.

Proposition C.3 (Certified value of omitted state). *Let x^* maximize W on X . Then*

$$J - K \geq \frac{1}{2} \text{dist}_Q(x^*, \mathcal{A})^2. \quad (40)$$

For any accepted $\hat{x}$ and valid $U_X \geq J$, put $\epsilon = U_X - W(\hat{x})$. The computable bound

$$J - K \geq \frac{1}{2} \left[\text{dist}_Q(\hat{x}, \mathcal{A}) - \sqrt{2\epsilon} \right]_+^2 \quad (41)$$

is valid. If Q is diagonal with entries $d_n > 0$ and $\mathcal{A}$ only pools decisions in cells C , the squared distance is $\sum_C \sum_{n \in C} d_n (\hat{x}_n - \bar{x}_C)^2$, where $\bar{x}_C = \sum_{n \in C} d_n \hat{x}_n / \sum_{n \in C} d_n$.

Proof. Strong concavity and optimality on the convex set X imply $W(x^*) - W(y) \geq \|x^* - y\|_Q^2 / 2$ for every $y \in X$; the argument also holds at switching kinks by applying strong concavity to the segment from x^* to y and letting its length tend to zero. Minimize the distance over the larger set $\mathcal{A}$. The same inequality gives $\|x^* - \hat{x}\|_Q \leq \sqrt{2\epsilon}$. Distance to a closed affine set is 1-Lipschitz in that norm. The diagonal formula is the weighted least-squares projection. $\square$

Two exact examples isolate the omitted coordinates. In the first, a three-date tree has one root, two equiprobable date-1 regimes, and a common date-2 regime. Let $a = \beta = 1$, $x \in [0, 1]$, stage reward $r_n x - x^2/2$, and reward coefficients $(0, 1/4, -1/4, 0, 0)$ in root, left, right, left-terminal, right-terminal order. Set root promise 2, date-1 caps $3/4$ and $5/4$, and terminal caps 1. The optimal accepted policy is $(1, 1/2, 1/2, 1/4, 3/4)$, with value $-25/32$. A date-regime table must pool the terminal actions and has optimum $(1, 1/4, 3/4, 1/2, 1/2)$, with value $-27/32$. The gain is $1/16$. The full optimizer reaches the two terminal histories with the same public regime and the same inherited tier $1/2$, but different remaining promises $1/4$ and $3/4$. Thus inherited tier cannot substitute for the promise in these continuation decisions. The curvature lower bound is $1/32$.

Conversely, consider two remaining dates, one regime, the same remaining promise 1, and inherited tiers 0 and 1 occurring with equal probability. With quadratic cost $(x^2 + (1 - x)^2)/2$, incoming switching cost $3|x - q|/2$, and final switching cost $|1 - 2x|/2$, the informed first actions are $1/4$ and $3/4$. Their average value is $-15/16$. A tier-blind common first action is optimally $1/2$, with value -1 . Thus the promise does not replace the inherited tier either. EC Section EC.11 proves both examples, explains the interpolation residual bound, and specifies the independent continuous-state checks.

C.3 A shared-table bridge for restriction values

A restriction on histories is not automatically a restriction that a Bellman recursion may impose independently in each successor. In particular, a time-only table must be the *same table* in all histories. Let a finite public memory m update by $m' = F_t(m, z')$, and let $u = (u_{t,m}) \in [0, 1]^K$ be a table selected once, before uncertainty is resolved. For positive weights $\omega_{t,m}$ and release radius $\delta \geq 0$, impose

$$|x_h - u_{t,m(h)}| \leq \delta \omega_{t,m(h)}. \quad (42)$$

At zero release this is exact history pooling. Eliminating the scalar table entry gives all pairwise inequalities $|x_h - x_{h'}| \leq 2\delta \omega_{t,m}$ within a cell. The factor two matters. Thus this institution is a subclass of Section A, not a new, incomparable benchmark.

For the remainder of this subsection use the positive-payment scalar Markov institution and the stage reward $g_t(z, q, x) = r_t(z)x - h_t(z)x^2/2 - \ell_t(z)|x - q|$, with $h_t(z) > 0$. Write $\mathcal{V}_t^m(z, q, b; u, \delta)$ for recursion (33) with memory appended and corridor (42) imposed. The global table is a read-only argument of every child problem. The restricted optimum is

$$V_M(\delta) = \max_{u \in [0,1]^K} \mathcal{V}_0^{m_0}(z_0, q_0, b_0; u, \delta). \quad (43)$$

Only tables admitting the specified root promise have finite value. Assume at least one such table exists at zero release.

Theorem C.4 (Shared-table, continuation-price bridge). *The value in (43) equals the optimized history-tree problem with (42). A maximizing contract has state (t, z, m, q, b) and one precommitted table. The value is nondecreasing, concave, and piecewise quadratic in δ . It need not equal the recursion that reselects the table in each child.*

Moreover, finite families of recursively certified affine planes

$$H_t(q, b; u, \delta) = A_t q + B_t b + \gamma_t^\top u + k_t \delta + C_t$$

give an upper bound on the optimized restricted comparator through the Benders-style root linear program

$$U_M(\delta) = \max_{u \in [0,1]^K, v} \{v : v \leq H_{0,a}(q_0, b_0; u, \delta) \text{ for every stored } a\}. \quad (44)$$

Every feasible policy with one shared table gives a lower bound $L_M(\delta)$. Consequently,

$$L_M(\delta) - U_M(0) \leq V_M(\delta) - V_M(0) \leq U_M(\delta) - L_M(0). \quad (45)$$

These certificates use state planes, policy witnesses, and a $K+1$ variable master problem, not a query history vector.

For finite horizons the class of recursively generated planes is extensive-form-dual representable: at any feasible fixed root promise and release, there exists a plane for which maximizing over the table gives $V_M(\delta)$ exactly. This is an existence result, not a polynomial bound on the number of planes needed to discover it. A root plane's release coefficient is the expected discounted sum of its local corridor prices. When a tight plane is also maximized by the optimizing table, this coefficient is a supergradient of V_M ; at a differentiability point it is its marginal release value.

Here is the constructive price recursion. For successor j , choose a valid plane with coefficients $(A_j, B_j, \gamma_j, k_j, C_j)$, or a convex combination of such planes. Put $p_j = P_t(z, j)$, $c = (t, m)$, and let I_j be the unrestricted feasible child-promise interval from Theorem C.2. Choose payment price $\eta \in \mathbb{R}$, participation price $\chi \geq 0$, corridor prices $\rho^+, \rho^- \geq 0$, and switching tension $|s| \leq \ell_t(z)$. With $a = a_t(z)$, $r = r_t(z)$,

$h = h_t(z)$, and $\bar{b} = \bar{b}_t(z)$, define

$$\begin{aligned}
 d &= r - s - a\eta - \rho^+ + \rho^- + \beta \sum_j p_j A_j, \\
 A &= s, \quad B = \eta - \chi, \\
 \gamma &= (\rho^+ - \rho^-)e_c + \beta \sum_j p_j \gamma_j, \\
 k &= \omega_c(\rho^+ + \rho^-) + \beta \sum_j p_j k_j, \\
 C &= \psi_h(d) + \chi \bar{b} + \beta \sum_j p_j \left[C_j + \max_{v \in I_j} (B_j - \eta)v \right], \quad \psi_h(d) = \max_{0 \leq x \leq 1} (dx - hx^2/2).
 \end{aligned} \tag{46}$$

Terminal coefficients are zero and the terminal promise is zero. An empty unrestricted child domain makes the parent infeasible. The use of unrestricted intervals keeps the plane valid as the shared table changes. Using table-dependent intervals without accounting for their dependence would not establish this affine certificate.

Proof outline. For a fixed table, conditioning and concatenation prove both directions of the state equivalence, without changing the table. Maximizing once over that table proves (43). The scalar interval-intersection criterion eliminates the table and identifies the pairwise release problem. Its value properties follow from Theorem A.1. For the upper bound, insert the child planes, price the exact payment equation, and add the nonnegative slacks of the participation and two corridor inequalities. The switching inequality is $-\ell|x - q| \leq -s(x - q)$. Maximizing the remaining scalar quadratic and each child promise gives (46); induction and the master problem give (45). To prove completeness, take an optimal tree primal-dual pair, normalize each price by its discounted reach probability, and construct the planes backward. Child payment prices then satisfy $\eta_j - \chi_j = \eta$, every priced slack vanishes, and the local box maximum is attained. Table stationarity makes the optimal shared table maximize the root plane. EC Section EC.13 supplies the complete argument, the normalizations, and the limits of this representation. $\square$

The participation term is essential to this construction: conditional marginal payment prices accumulate along a path as $\eta_j = \eta + \chi_j$. This is the nested-continuation counterpart of $A^\top \mu$ in Proposition 3.1. The coefficient γ keeps the *cross-history* table stationarity, while k keeps the release rent. These two conditions are absent from an unrestricted promise recursion. General signed or nonadditive constraints in (3) do not automatically admit this scalar bridge.

An exact two-period example makes the distinction observable. Set $\beta = 1$, equal child probabilities, all payment coefficients and quadratic curvatures to one, rewards $(1, 0, 2)$, root promise one, and child caps $(2/5, 4/5)$. A shared time-only child tier gives $(x_0, x_L, x_H) = (3/5, 2/5, 2/5)$ and value $37/50$. With no pooling the optimum is $(3/5, 0, 4/5)$ and value $53/50$. Reselecting a separate table in each child falsely reports the latter value even at zero release. Correct release has events at $\delta = 1/10, 3/10, 2/5$; the exact policies, recursively reconstructed prices, and supporting upper planes are independently replayed in EC Section EC.14.

C.4 Choosing an implementable architecture

The hierarchy becomes an endogenous decision when an architecture j has a declared installation and memory cost and a release cost $C_j(\delta)$. For example, $C_j(\delta) = f_j + c_j T |M_j| + a_j \delta + b_j \delta^2 / 2$, with $b_j \geq 0$, prices a maintained table and the permitted organizational discretion. These are specified design inputs, not estimated causal values of information. The provider chooses

$$\max_{j, 0 \leq \delta \leq \bar{\delta}_j} \{V_{M_j}(\delta) - C_j(\delta)\}.$$

On a regular release segment an interior optimum equates the recursively priced release rent to $C'_j(\delta)$; segment endpoints must also be compared. For a finite offered design menu, bounds L_j, U_j certify the selected feasible design within ε whenever $\max_i (U_i - C_i) - (L_j - C_j) \leq \varepsilon$. Replacing a zero-release contract by full flexibility in the example is worthwhile precisely when its incremental installation cost is below $8/25$. For a linear release cost $C(\delta) = \delta$, the net optimum is $\delta = 1/15$, rather than full flexibility. This decision comparison uses optimized continuation-feasible contracts on both sides.

D Implemented Gains and Reusable Comparator Certificates

A feasible implemented $\hat{x}$ and a valid upper bound $U_Y \geq K$ give the economic certificate

$$W(\hat{x}) - K \geq W(\hat{x}) - U_Y. \quad (47)$$

An upper bound $U_X \geq J$ separately gives $J - W(\hat{x}) \leq U_X - W(\hat{x})$. The first establishes an improvement over an optimized restriction; the second establishes implementation accuracy. Neither follows from predictor error or an optimizer's termination label. EC Section EC.7 states the box-conjugate bound used by the independent verifier, retaining every participation row, equality, and switching term.

D.1 A comparator envelope requiring no new query solve

Consider a fixed restricted geometry Y and fixed convex cost ϕ with strong-convexity matrix $D_0 \succ 0$:

$$K(h) = \max_{x \in Y} \{(b + U^\top h)^\top x - \phi(x)\}.$$

Let the columns of Z_Y span a linear space containing every difference of two points of Y , and choose

$$L_Y \geq \|UZ_Y(Z_Y^\top D_0 Z_Y)^{-1/2}\|^2.$$

A trace bound on the corresponding positive semidefinite matrix is sufficient and can be evaluated rationally.

Proposition D.1 (Certified reusable restricted-value envelope). *At an anchor h_i , let $z_i \in Y$ and $U_i \geq K(h_i)$ be certified, and put $\delta_i = U_i - W_{h_i}(z_i)$. Then, for every reward query h ,*

$$K(h) \leq U_i + (U z_i)^\top (h - h_i) + \sqrt{2L_Y \delta_i} \|h - h_i\| + \frac{1}{2} L_Y \|h - h_i\|^2. \quad (48)$$

The minimum of these bounds over a finite cache remains valid. After storing U_i , Uz_i , δ_i , and h_i , each query uses order r arithmetic operations per anchor, where r is the reward dimension, and no restricted optimization. The bound is valid even when the anchor decision is not exactly optimal.

Proof. Strong convexity gives $\|z_i - x^*(h_i)\|_{D_0}^2 \leq 2\delta_i$. Restricting differences to the displayed linear space bounds the gradient error by $\|U(z_i - x^*)\| \leq \sqrt{2L_Y\delta_i}$. The same strong-monotonicity argument gives an L_Y -Lipschitz gradient $\nabla K(h) = Ux^*(h)$. Apply the smooth upper Taylor bound, substitute $U_i \geq K(h_i)$, and bound the anchor-gradient error by Cauchy–Schwarz. EC Section EC.8 gives the full argument, including exact outward square-root rounding. $\square$

This is a nearby-reward reuse mechanism, not a theorem for an arbitrarily changing accepted geometry. Changing continuation matrices, budgets, switching coefficients, or information restrictions requires a new valid cache or a bound that explicitly covers those changes. A loose envelope is still valid but may not certify a positive economic gain. A deployment can then request a new comparator certificate; that solve must be charged. Accepted-policy implementation itself need not perform an economic-comparator batch on every query. Theorem D.3 supplies a separate valid bound for jointly changing constraints and friction.

D.2 Why the robust outer comparator is locally continuous

Write a designed family as

$$Y_\theta(\rho) = \{x \in C_0 : (A^0 + \rho A_\theta)x \leq b^0 + \rho b_\theta\},$$

where C_0 is the common compact box and equality-restricted class, and θ belongs to a compact uncertainty set. Let η_i bound $\sup_{\theta, x \in C_0} |(A_\theta)_i x - (b_\theta)_i|$ and put $O(\rho) = \{x \in C_0 : A_i^0 x \leq b_i^0 + \rho \eta_i\}$. This radius-dependent outer class contains every $Y_\theta(\rho)$ and equals the nominal class at zero. It is not a fixed large union bound switched on at every positive radius.

Proposition D.2 (Local outer-comparator loss). *Suppose an anchor in C_0 has slack at least $\gamma_i > 0$ in every perturbed row for $0 \leq \rho \leq \rho_0$. Suppose the objective W_θ is uniformly L_W -Lipschitz on C_0 , whose diameter is D_C . Then*

$$0 \leq \max_{O(\rho)} W_\theta - \max_{Y_\theta(\rho)} W_\theta \leq L_W D_C \max_i \frac{2\rho\eta_i}{\gamma_i + 2\rho\eta_i}. \quad (49)$$

Thus the outer-set component of certificate slack vanishes at least linearly as the radius tends to zero under these assumptions. The constant can be large when an important accepted row has little strict slack.

Proof. An outer-feasible decision violates any true row by at most $2\rho\eta_i$. Mix it with the anchor by the largest fraction in (49). Every resulting true row is feasible, and the common equalities and box remain feasible by convexity. The movement is at most that fraction times D_C . Apply the Lipschitz bound to an outer maximizer. $\square$

The row envelope can be tightened by taking its support over the equality-restricted box rather than an unrestricted box. EC Section EC.9 also gives a valid successive-outer-set refinement. Neither a small

uncertainty radius nor set containment alone establishes field robustness: the uncertainty set remains a designed, uncalibrated family. Without a uniform anchor, the linear rate above is not asserted. The finer-radius experiment distinguishes high local sensitivity from an actual positive-radius discontinuity.

D.3 Jointly changing commitments and switching costs

The reward-only envelope does not address the case in which the same operational shock changes revenue, continuation coefficients, capacity, and switching friction. Replacing these correlated constraints by independent row envelopes loses contractual information. We instead retain the actual coefficient vector in a reusable dual certificate.

For this construction, write $Q = H + R^\top R$, where $H \succ 0$ is diagonal, and combine the payment and information equalities into $F_\theta x = f_\theta$. The restricted problem is

$$\begin{aligned} K_\theta &= \max_{x \in Y_\theta} W_\theta(x), \quad Y_\theta = \{x \in [\ell, u] : A_\theta x \leq b_\theta, F_\theta x = f_\theta\}, \\ W_\theta(x) &= c_\theta + p_\theta^\top x - \frac{1}{2} x^\top H x - \frac{1}{2} \|R x\|^2 - \lambda_\theta \sum_j \kappa_j |(Dx - h)_j|. \end{aligned} \quad (50)$$

The common box, H, R, D, h , and positive weights κ are fixed; all displayed θ -dependent coefficients may change together, with $\lambda_\theta \geq 0$. The known constant c_θ permits gain-relative objectives. Each restricted class is nonempty. Define

$$\psi_i(d) = \max_{\ell_i \leq v \leq u_i} \{dv - H_{ii}v^2/2\}, \quad v_i(d) = \min\{u_i, \max\{\ell_i, d/H_{ii}\}\}.$$

A cached price $a = (\alpha, \mu, \nu, \tau)$ has $\mu \geq 0$ and $|\tau_j| \leq \kappa_j$. Its switching tension at a query is $\lambda_\theta \tau$, not the unscaled tension from the training problem.

Theorem D.3 (Correlated comparator reuse and its local loss). *For any such cached price, put*

$$d_\theta(a) = p_\theta - R^\top \alpha - A_\theta^\top \mu - F_\theta^\top \nu - \lambda_\theta D^\top \tau, \quad (51)$$

$$\mathcal{U}_\theta(a) = c_\theta + \frac{1}{2} \|\alpha\|^2 + \mu^\top b_\theta + \nu^\top f_\theta + \lambda_\theta \tau^\top h + \sum_i \psi_i(d_{\theta,i}(a)). \quad (52)$$

Then $K_\theta \leq \mathcal{U}_\theta(a)$ for every query. The minimum over a finite cache remains valid without a new restricted optimization, an interior feasible anchor, or an unchanged active set.

More precisely, for every feasible restricted z , the certificate gap is the sum of four nonnegative terms:

$$\begin{aligned} \mathcal{U}_\theta(a) - W_\theta(z) &= \frac{1}{2} \|\alpha - Rz\|^2 + \sum_i \left\{ \psi_i(d_{\theta,i}) - d_{\theta,i} z_i + \frac{1}{2} H_{ii} z_i^2 \right\} \\ &\quad + \mu^\top (b_\theta - A_\theta z) + \lambda_\theta \sum_j \left\{ \kappa_j |(Dz - h)_j| - \tau_j (Dz - h)_j \right\}. \end{aligned} \quad (53)$$

Thus the loss separates into resource-price error, coordinate-response error, unused priced continuation capacity, and switching-price error.

For a local statement, let z_0 be the exact restricted optimizer at θ_0 , let the cached price attain equality there, and set $v_0 = d_{\theta_0}(a) - Hz_0 \in N_{[\ell, u]}(z_0)$. Suppose the query optimizer z_θ stays on the priced face: $\mu^\top(b_\theta - A_\theta z_\theta) = 0$, $v_0 \in N_{[\ell, u]}(z_\theta)$, and every switching term in (53) vanishes. With $e_\theta = d_\theta(a) - Hz_\theta - v_0$, one has

$$0 \leq \mathcal{U}_\theta(a) - K_\theta \leq \frac{1}{2} \|R(z_\theta - z_0)\|^2 + \frac{1}{2} e_\theta^\top H^{-1} e_\theta. \quad (54)$$

In particular, locally Lipschitz coefficients and a locally Lipschitz optimizer give a quadratic reuse loss on this priced face. The global upper certificate does not require these local conditions.

Proof. Linearize the resource square with α , lower-bound each absolute value by its normalized tension, and subtract the feasible inequality and equality residuals. Maximizing the remaining separate concave quadratics on the box gives (52). Adding and subtracting their values at z gives (53); equality feasibility removes the equality-price residual. At an exact anchor the resource term implies $\alpha = Rz_0$, and each coordinate is a box maximizer, giving $v_0 \in N_{[\ell, u]}(z_0)$. On the priced face only the first two terms remain. For a box-feasible displacement $y - z_\theta$, the second term is the supremum of $(v_0 + e_\theta)^\top (y - z_\theta) - \|y - z_\theta\|_H^2 / 2$. The normal term is nonpositive; the unrestricted supremum of the remainder is $e_\theta^\top H^{-1} e_\theta / 2$. This proves (54). EC Section EC.12 gives explicit constants and the boundary cases. $\square$

The continuation prices need not be refitted to remain valid. They may, however, become expensive prices for slack that the query optimizer no longer uses. This distinction explains both the benefit of reuse and its failure mode. With affine coefficient dependence in d parameters, cached matrix-price products reduce evaluation to order $n(d + 1)$ arithmetic operations per anchor; exact rational bit cost and all anchor construction remain additional work. This is not the dimension-free reward-only bound of Proposition D.1, and it is not a measured speedup.

The quadratic conclusion must not be extended across an arbitrary degenerate face. For $|t| \leq 1/4$, the problem $\max\{x - x^2/2 : -1/2 \leq x \leq 1/2, x \leq t, x \leq -t\}$ has the following behavior. An exact price $(1, 0)$ at $t = 0$ gives $\mathcal{U}_t = t$, whereas $K_t = -|t| - t^2/2$. The reuse gap is $2t + t^2/2$ for $t > 0$. The released first constraint has positive cached price and positive slack. Numerical anchor error likewise does not disappear by declaring an anchor exact: (52) remains valid, but the local quadratic assertion requires the exact-anchor assumptions.

Corollary D.4 (Uniform certification with correlated uncertainty). *Suppose all θ -dependent coefficients in (50) are affine on a compact polytope Θ , and $\lambda_\theta \geq 0$ there. Partition Θ into polytopes C with vertex sets V_C . A fixed implemented policy $\hat{x}$ is feasible for the full accepted class throughout Θ if it satisfies its affine inequalities and equalities at all vertices. For any finite cache $\mathcal{A}$,*

$$\inf_{\theta \in \Theta} \{W_\theta(\hat{x}) - K_\theta\} \geq \min_C \max_{a \in \mathcal{A}} \min_{v \in V_C} \{W_v(\hat{x}) - \mathcal{U}_v(a)\}. \quad (55)$$

Subdivision of cells cannot decrease this lower certificate, and adding cached prices cannot decrease it. Neither operation requires new comparator solves.

Proof. The function $\mathcal{U}_\theta(a)$ is convex in θ : its residuals are affine and each ψ_i is convex. The fixed-policy reward is affine, so their difference is concave and is bounded below on a cell by its minimum vertex value.

Choose the best fixed anchor on that cell, then take the worst cell. Convexity also proves the subdivision assertion; retaining old prices proves the cache assertion. $\square$

The anchor must remain fixed inside the vertex minimum in (55). Taking the best anchor separately at vertices and interpolating those best values is generally invalid. The EC gives an exact failure example and a positive-gain example: four cells certify at least 0.04919921 uniformly against an optimized restricted contract under jointly changing reward, friction, and participation coefficients. No sampled parameter is substituted for a continuous uncertainty guarantee. The set is designed, not statistically calibrated.

D.4 A certified cache-governance rule

The global bound in Theorem D.3 remains valid when its favorable priced face changes. Its quadratic-loss clause is not a license to trust a stale cache. The following rule uses only audited feasible witnesses and the actual query coefficients.

Proposition D.5 (Quality and refresh certificate). *Let $U = \min_a U_\theta(a)$ be the cached comparator upper bound, and let $z^w \in Y_\theta$ be any independently audited restricted witness. Then*

$$0 \leq U - K_\theta \leq U - W_\theta(z^w).$$

Thus a computed width at most ε certifies cache quality without knowing the query optimizer. Otherwise the controller must refresh the price/witness pair or declare the requested tolerance uncertified. Failure of this sufficient test does not prove that the cache is inaccurate. Deployment additionally requires an audited full-class policy and $W_\theta(\widehat{x}) - U$ above the required net improvement.

The width is the exact four-term decomposition in Theorem D.3 at z^w . A repaired cached table or a feasible state policy from Theorem C.4 can provide the witness; a repair is not credited as a fresh optimized comparator. No empirical price is interpreted as a directional derivative at degeneracy.

A stronger face test is also available. Store an independent active equality basis, its switching signs, and the exact-anchor prices. At a query, solve only the equality-constrained KKT linear system on that face, then verify *all* primal inequalities, dual signs, switching tensions, and complementarity against the query model. If these checks pass, convex optimality certifies the query restricted optimum; the additional cached-face conditions in Theorem D.3 then certify its quadratic bound. A nonsingular factorization can be reused only when its coefficient matrix is unchanged. Otherwise factorization and verification are additional costs, not constant-time cache lookups. EC Section EC.15 gives the system and the complete accounting.

D.5 Uniform certificates for a declared adaptive policy

The preceding uniform corollary concerns a fixed policy. To certify an actual adaptive deployment map, partition its parameter domain into simplices and declare $\widehat{x}(\theta)$ affine on each simplex. Assume every restricted query set is nonempty, the common box and quadratic/switching geometry stay fixed, and $A_\theta, b_\theta, F_\theta, f_\theta, p_\theta, c_\theta, \lambda_\theta$ are affine, with $\lambda_\theta \geq 0$. Write $E_\theta x = e_\theta$ for the full-class equalities; $F_\theta x = f_\theta$

additionally includes the comparator restriction. The accepted equalities are affine too. The deployed policy is audited against $(A_\theta, b_\theta, E_\theta, e_\theta)$, not against comparator-only pooling rows. Feasibility must hold throughout each cell, not only at sampled queries.

Proposition D.6 (Adaptive, correlation-preserving cell certificate). *Let the vertices of a simplex be v_i , with policy values $x_i = \widehat{x}(v_i)$. A sufficient condition for $A_\theta \widehat{x}(\theta) \leq b_\theta$ throughout the simplex is componentwise nonnegativity of all coefficients*

$$B_{ii} = b_i - A_i x_i, \quad B_{ij} = \frac{1}{2}(b_i + b_j - A_i x_j - A_j x_i) \quad (i < j). \quad (56)$$

Apply the same formula to the accepted equalities (E_θ, e_θ) and require every coefficient to be zero; box feasibility is vertex-checkable. Refine cells at switching-sign hyperplanes. For a cached price a , define the quadratic polynomial

$$f_a(\theta) = W_\theta(\widehat{x}(\theta)) - \sum_i \lambda_i(\theta) U_{v_i}(a),$$

where the $\lambda_i(\theta)$ in this display are barycentric weights, not friction scales. Its quadratic Bernstein coefficients are

$$g_{ii,a} = f_a(v_i), \quad g_{ij,a} = 2f_a((v_i + v_j)/2) - \frac{1}{2}[f_a(v_i) + f_a(v_j)].$$

If every cell passes the feasibility test, a uniform gain lower bound is

$$\inf_{\theta} \{W_\theta(\widehat{x}(\theta)) - K_\theta\} \geq \min_{\text{cells}} \max_a \min_{i \leq j} g_{ij,a}. \quad (57)$$

The guarantee includes the interior of every cell and the prescribed boundary policy. Discontinuous piecewise policies require a declared boundary assignment and a certificate for each assigned piece.

The proof expands the correlated products in the nonnegative Bernstein basis. Convexity of $U_\theta(a)$ puts it below its vertex chord; the Bernstein minimum bounds the remaining quadratic. Checking only vertices of $A_\theta \widehat{x}(\theta)$ is false in general. For example, $A_\theta = \theta$, $\widehat{x}(\theta) = 1 - \theta$, $b_\theta = 0$ passes both endpoint checks but violates the constraint in the interior. The off-diagonal coefficient in (56) is $-1/2$ and rejects it. EC Section EC.16 proves the result and gives a nonconstant, uniformly feasible policy with certified gain 9/1024 under jointly changing rewards, a continuation coefficient, and friction scale.

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Table 1: Exact computation on coupled recombining graphs.

Design	Vertices	Knots	Segments	Price pairs	Solve (s)	Audit (s)
8 dates, 2 states	15	30	180	33	0.007	0.018
16 dates, 2 states	31	62	699	101	0.030	0.078
32 dates, 2 states	63	129	2801	69	0.117	0.336
64 dates, 2 states	127	246	8883	1439	0.643	1.633
16 dates, 3 states	46	96	1677	172	0.091	0.231
32 dates, 3 states	94	180	5405	337	0.285	0.807
64 dates, 3 states	190	332	20452	580	1.190	3.512
32 dates, 2 states*	63	146	2859	273	0.123	0.359
64 dates, 2 states*	127	278	11094	559	0.566	1.583

*Fixed shared time-only table with positive corridor release. Knots count the global union, segments count all stored response pieces, and price pairs count reached public/incoming-price states. Every value has an exact system-level rational primal–dual equality. Times are single-run platform observations, not replicated speed estimates; audit is independent of the generator.

Table 2: The exact memory frontier for eight renewal regimes.

Symbols	Writable bits	Optimal reward	Loss from full information
1	0	35/324	119/162
2	1	61/108	5/18
3	2	56/81	49/324
4	2	245/324	7/81
5	3	169/216	13/216
6	3	29/36	1/27
7	3	535/648	11/648
8	3	91/108	0

Caps are $b_j = j/9$ and the exact root promise is $1/2$. Bits count an index into a stored decision table, not the read-only table or numeric storage. Two bits permit four symbols; unused capacity is not required. No memory costs are calibrated.